Demonstration of Proposed Features

Date	02 July 2026
Team ID	
Project Name	Rising Waters – A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated, serving as evidence of the team's ability to deliver on the proposed solution.

S. No	Feature Name / Description	Status	Demonstrated	Remarks
1	Historical rainfall EDA (national trend, seasonality, regional comparison)	Implemented	Yes	Delivered in the notebook with distribution, heatmap, trend and regional charts
2	Multi-model training & comparison (Decision Tree, RF, KNN, XGBoost)	Implemented	Yes	All four models trained/evaluated with confusion matrices & classification reports
3	Automatic best-model selection & persistence	Implemented	Yes	XGBoost selected (95.65% accuracy) and bundled with scaler + feature names via joblib
4	Home page with live model accuracy & overview	Implemented	Yes	Stats pulled live from the loaded model bundle via a Flask context processor
5	Guided, grouped prediction form	Implemented	Yes	10 fields grouped into 3 sections with labels, units, help text and defaults
6	Flood Chance / No Flood Chance result pages	Implemented	Yes	Dedicated templates render probability and next-step guidance based on prediction
7	Multi-target deployment (Docker / Cloud Foundry / Procfile)	Implemented	Yes	All three deployment configs included and demonstrated locally via Docker

Feature Implementation Summary

Metric	Value
Total Features Proposed	7
Total Features Implemented	7
Total Features Demonstrated	7